Coproduct transducers for harmony

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At a glance

- **ISL functions** (Chandlee, 2014) give us great empirical coverage for local phenomena.
- Long-distance phenomena like **vowel harmony** or **long-distance {a/dis}similation** are computed with **TSL functions** (Burness et al., 2021)
- **MTSL functions** (Burness, 2022) are useful for modeling possible interactions between multiple long-distance processes.
- This presentation introduces **coproduct transducers**, a highly parameterized model that carves up the class of rational string functions.
- Several long-distance processes are studied as transductions over a tier-coproduct.

Roadmap

- Briefly talk intuitions for MTSL functions and coproduct transductions
- Sketch out how to set up coproduct transducers for some vowel harmony patterns discussed in the MTSL literature (Burness & McMullin, 2020; Burness, 2022; Burness et al., 2024)
- Explore how we can tweak the parameters to generate more complex maps, like those understood to be *weakly deterministic* (Heinz and Lai, 2013) and *unbounded circumambient* (Jardine, 2016)

Uyghur – Last resort triggers

- Suffixal vowels agree in backness with suitable vowel to their left
  - /i e/ are transparent
- Dorsal consonants serve as the harmony source only if no suitable vowel
  - Velars cause –back
  - Uvulars cause + back

Harmony with dorsal

(a) gezit-tæ ‘newspaper-LOC’

(b) qiʁ’iz-da ‘Kyrgyz-LOC’

Harmony across dorsal

(c) rak-ta ‘shrimp-LOC’

(d) mæʃq-tæ ‘exercise-LOC’

MTSL Functions (*à la* Burness)

Uyghur harmony is computable with an MTSL function that tracks one superset tier of vowels and dorsals, and one subset tier of just vowels

Superset Tier = {y, u, ø, o, æ, a, k, g, q, ʁ}
Subset Tier = {y, u, ø, o, æ, a}

/rak-tA/ → [rakta]

Superset Window	→	λ	→	a	→	k	→	k	→	k
Subset Window	→	λ	→	a	→	a	→	a	→	a
Input	r		a		k		t		A	
Output	r		a		k		t		a	

/gezit-tA/ → [gezittæ]

Superset Window	→	g	→	g	→	g	→	g	→	g	→	g	→	g	→	g
Subset Window	→	λ	→	λ	→	λ	→	λ	→	λ	→	λ	→	λ	→	λ
Input	g		e		z		i		t		t		A			
Output	g		e		z		i		t		t		æ			

MTSL Functions (*à la* Burness)

However, we can set up strange MTSL maps by exploiting the tier windows:

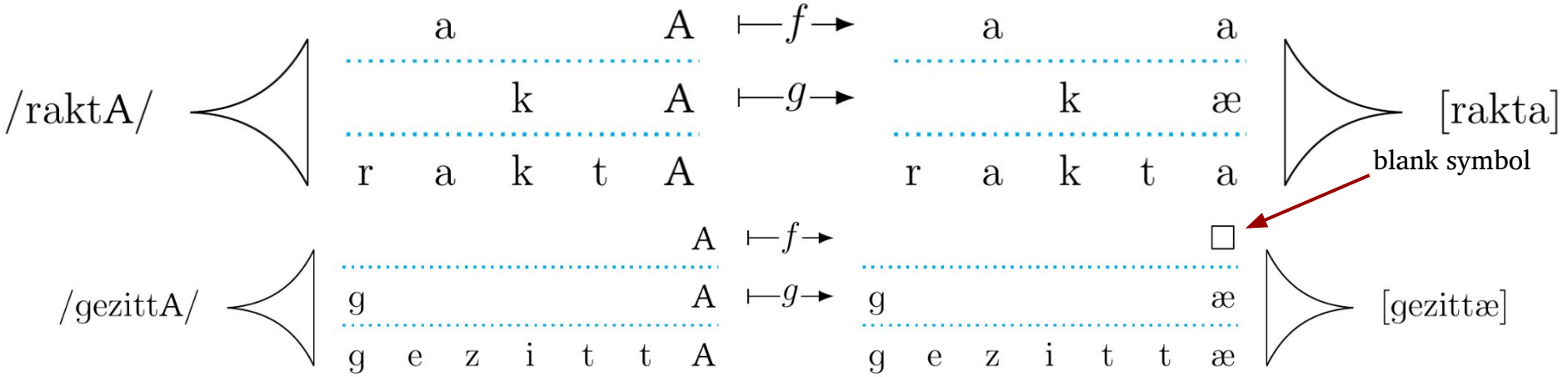
"*a* maps to *d* when preceded at any distance by both *b* and *c*, but *c* most recently"

Superset tier = {b,c}
Subset tier = {b}

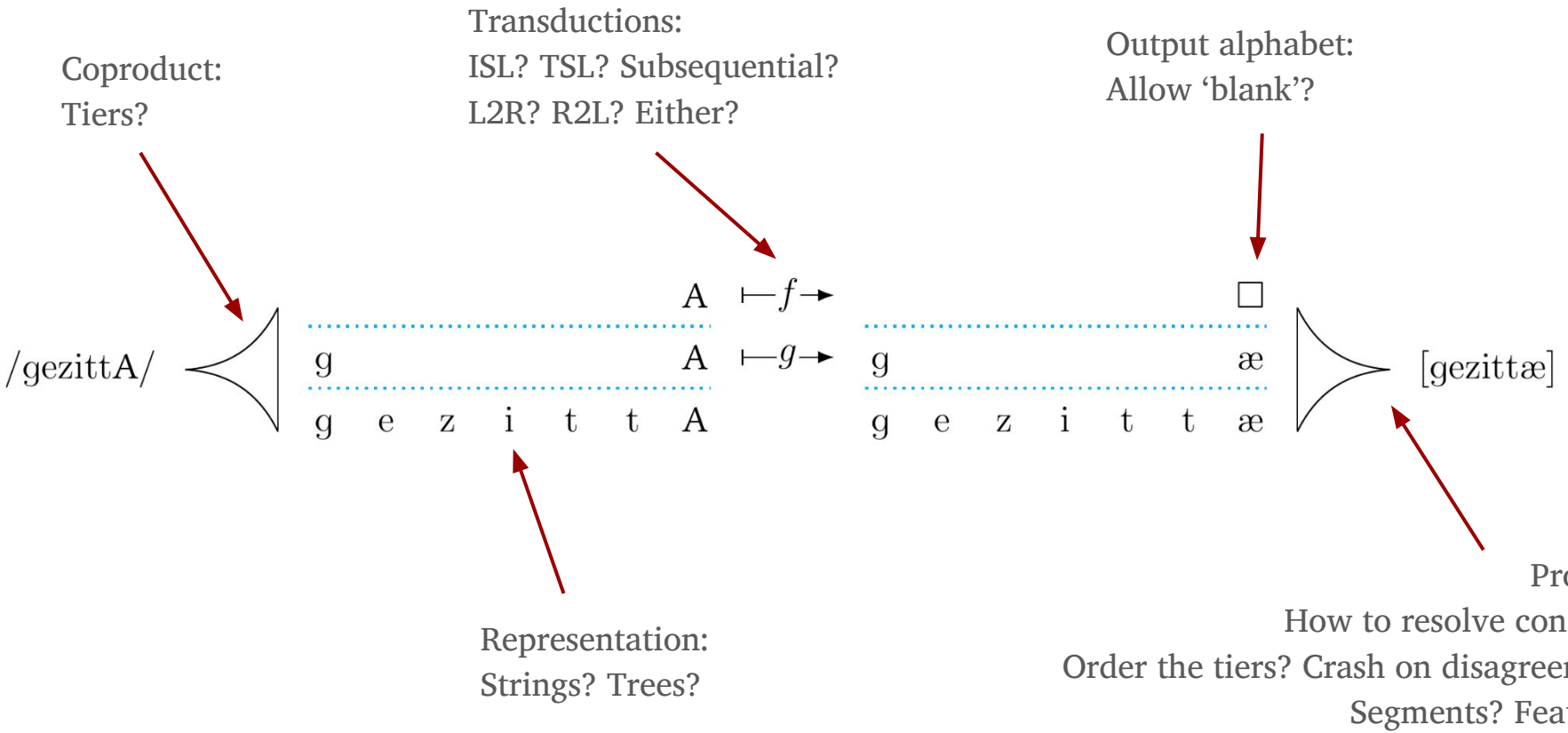
Superset Window	→	λ	→	b	→	b	→	c	→	c	→	b	→	b
Subset Window	→	λ	→	b	→	b	→	b	→	b	→	b	→	b
Input	a		b		a		c		a		b		a	
Output	a		b		a		c		d		b		a	

Intuition: Coproduct transduction

- Q: How can we ensure we get no interactions between tiers?
- A: Decompose the string into its tiers, run a transducer over each tier individually, and somehow unify their outputs.



The many parameters of coproduct transductions

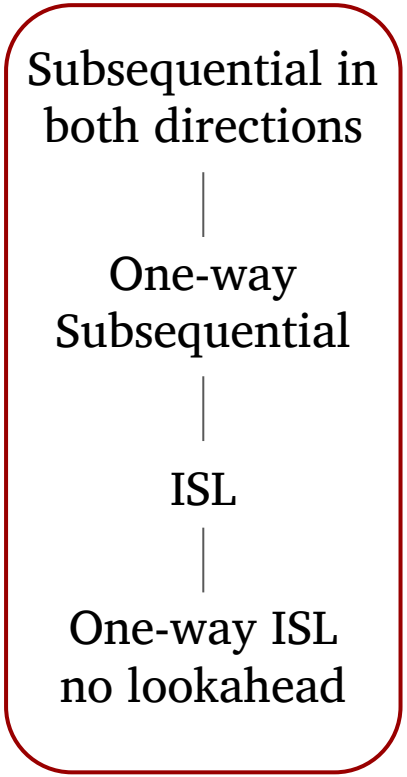


The many parameters of coproduct transductions

For our purposes here, we fix the following conventions:

Representation:	Fixed to strings
Coproduct:	Fixed to tier-decomposition
Output:	We allow blank symbol □
Product:	Conflicts are resolved by linear order

As for the power of the tier transductions: we see what settings are necessary to get vowel harmony working, and explore applications to more complex phonological maps outside the domain of vowel harmony



Uyghur – Last resort triggers

f		□			a							□
g				□		æ	□					æ
In	r	a	k	t	A	g	e	z	i	t	t	A
Out	r	a	k	t	a	g	e	z	i	t	t	æ

- We can represent the transformations in a more compact tabular form
- Both *f* and *g* are computed by L2R-ISL transducers without lookahead / delayed output

Subsequential in both directions

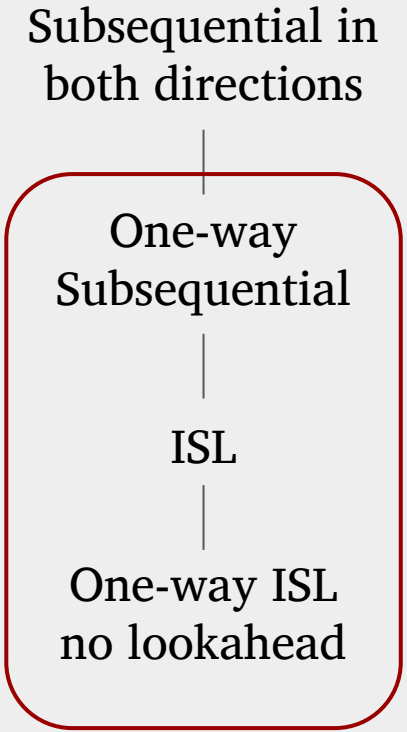
One-way Subsequential

ISL

One-way ISL no lookahead

Doing some harmony maps

What do we need to get vowel harmony working?



Turkish – Two harmony sources with one target

Backness harmony

- Suffixal vowels agree in backness with the vowel to their left
- Can agree instead with a subset of intervening consonants
  - +back: /k g l/
  - −back: /kʲ gʲ lʲ/

Roundness harmony

- Suffixal high vowels agree in rounding with the vowel to their left

One source

(a) ip-in ‘rope-GEN’
(b) sap-ɯn ‘stalk-GEN’
(c) jyz-yn ‘face-GEN’
(d) son-un ‘end-GEN’

Two sources

(e) usulʲ-y ‘system-ACC.SG’
(f) sualʲ-i ‘question-ACC.SG’

Turkish – Two harmony sources with one target

f								□	$\begin{bmatrix}-bk\\ \square rd\end{bmatrix}$
g		□		$\begin{bmatrix}+bk\\ -rd\end{bmatrix}$			□	□		$\begin{bmatrix}+bk\\ -rd\end{bmatrix}$
In	s	a	p	$\begin{bmatrix}\square bk\\ \square rd\end{bmatrix}$	n	s	u	a	l^{j}	$\begin{bmatrix}\square bk\\ \square rd\end{bmatrix}$
Out	s	a	p	$\begin{bmatrix}+bk\\ -rd\end{bmatrix}$	n	s	u	a	l^{j}	$\begin{bmatrix}-bk\\ -rd\end{bmatrix}$

- We just need to use an enriched alphabet that lets us do feature-wise conflict resolution
- Under that assumption, this has the same complexity as last resort harmony in Uyghur

Subsequential in both directions

One-way Subsequential

ISL

One-way ISL no lookahead

Tamashek Tuareg – Local process beating long-distance

g Labial dissimilation: /m/ in prefix becomes [n] if root contains labial consonant (besides [w])

(a) -m-VrtVj- ‘become mixed’

(b) -n-VkmVm- ‘be squeezed’

f Local nasal place assimilation: [n] becomes [m] before [b]

(c) -æm-bæbbɑ- ‘carried each other’

h Sibilant harmony: /s/ in prefix matches following sibilant in anteriority, voicing, and pharyngealization

(d) -s-VŋŋV- ‘cook’

(e) -s-VsVfVr- ‘treat (patient)’

(f) -sˤ-VsˤuhV- ‘strengthen’

(g) -ʃ-VluʃV- ‘clean sand from’

(h) -z-VjVzzV- ‘scrutinize’

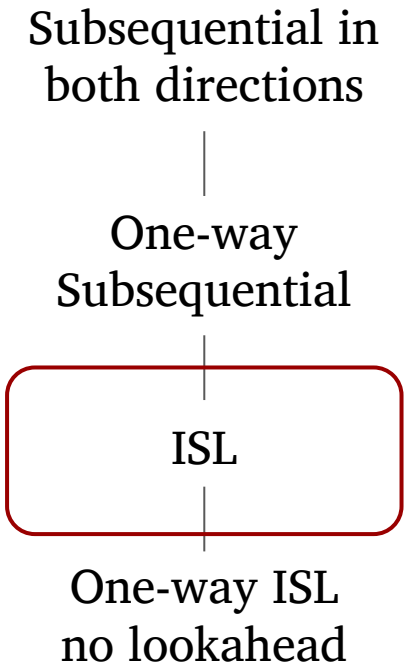
(i) ɑ-zˤ-ən:-ət-əlməzˤ ‘spitting saliva’

	g on its own						*f* overriding *g*							*g* and *h* applying simultaneously										
f	□	□	□	□	□	□	□	m	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
g	n			□		□		n	□		□	□					n					□		
h															zˤ									zˤ
In	m	V	k	m	V	m	æ	m	b	æ	b	b	ɑ	ɑ	s	ə	m	ə	t	ə	l	m	ə	zˤ
Out	n	V	k	m	V	m	æ	m	b	æ	b	b	ɑ	ɑ	zˤ	ə	n:	ə	t	ə	l	m	ə	zˤ

Tamashek Tuareg – Local process beating long-distance

	g on its own						*f* overriding *g*						
f	□	□	□	□	□	□	□	m	□	□	□	□	□
g	n			□		□		n	□		□	□	
h													
In	m	V	k	m	V	m	æ	m	b	æ	b	b	ɑ
Out	n	V	k	m	V	m	æ	m	b	æ	b	b	ɑ

- Following Burness et al. (2024), we factor out morphological conditions on labial dissimilation (occurs only on prefixes)
- Under that stipulation, this is ISL-2 over each tier



Samala – Long-distance beating local process

f Regressive sibilant harmony: Anteriority of rightmost sibilant spreads leftward

(a) /s-xalam-ʃ/ → [ʃxalamʃ] ‘it is wrapped’

g Regressive palatalization: /s/ palatalizes to [ʃ] before [t], [n], [l]

(b) /s-niʔ/ → [ʃniʔ] ‘his neck’

When both processes are triggered, the long-distance harmony beats out the local palatalization

(c) /s-net-us/ → [snetus] ‘he does it to him’

Samala – Long-distance beating local process

f	ʃ						ʃ		s					s
g	□	□	□	□	□	□	□		ʃ	□	□	□	□	□
In	s	x	a	l	a	m	ʃ		s	n	e	t	u	s
Out	ʃ	x	a	l	a	m	ʃ		s	n	e	t	u	s

- ISL isn't enough here. *f* is a Right 2-OSL function, becase there may be any number of sibilants in a Samala word
- Thus allowing one-way subsequential transdcers seems to increase our generative capacity further. How far can we push it?

Subsequential in both directions

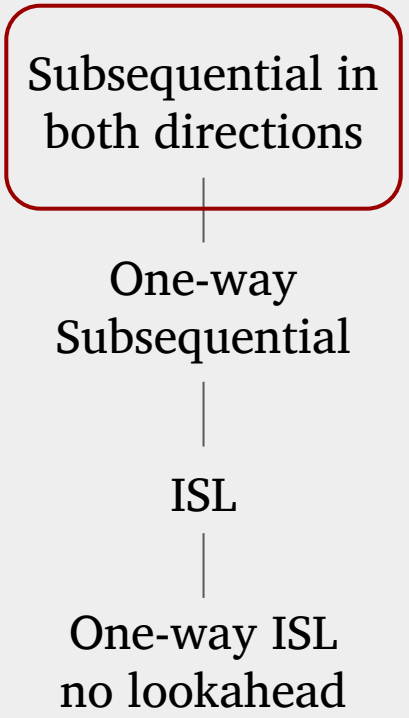
One-way Subsequential

ISL

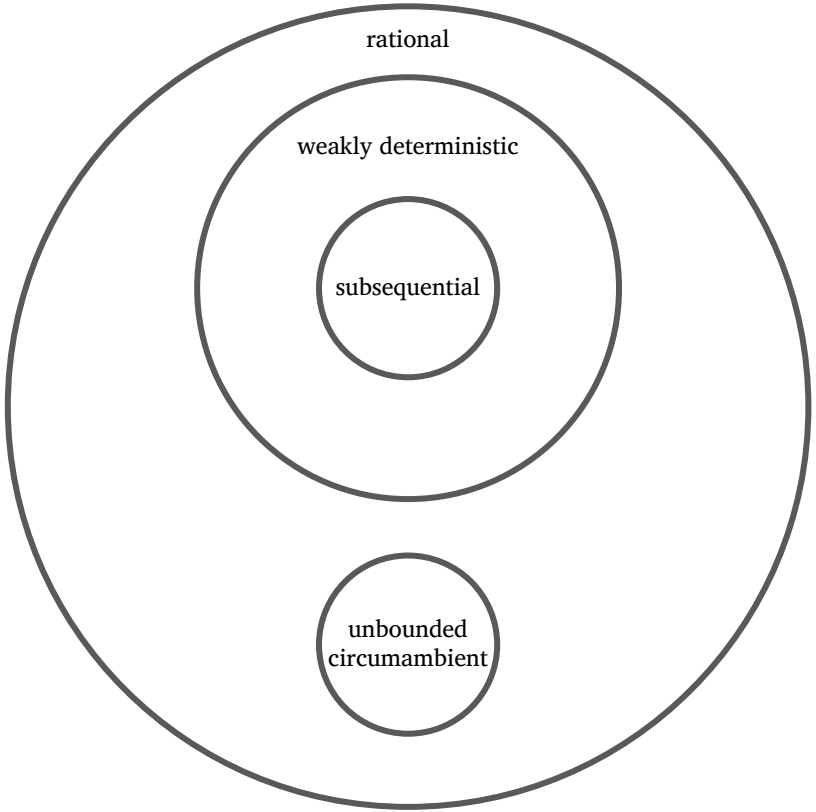
One-way ISL no lookahead

Doing more advanced maps

Weak determinism and unbounded circumambience



Weak determinism and unbounded circumambience



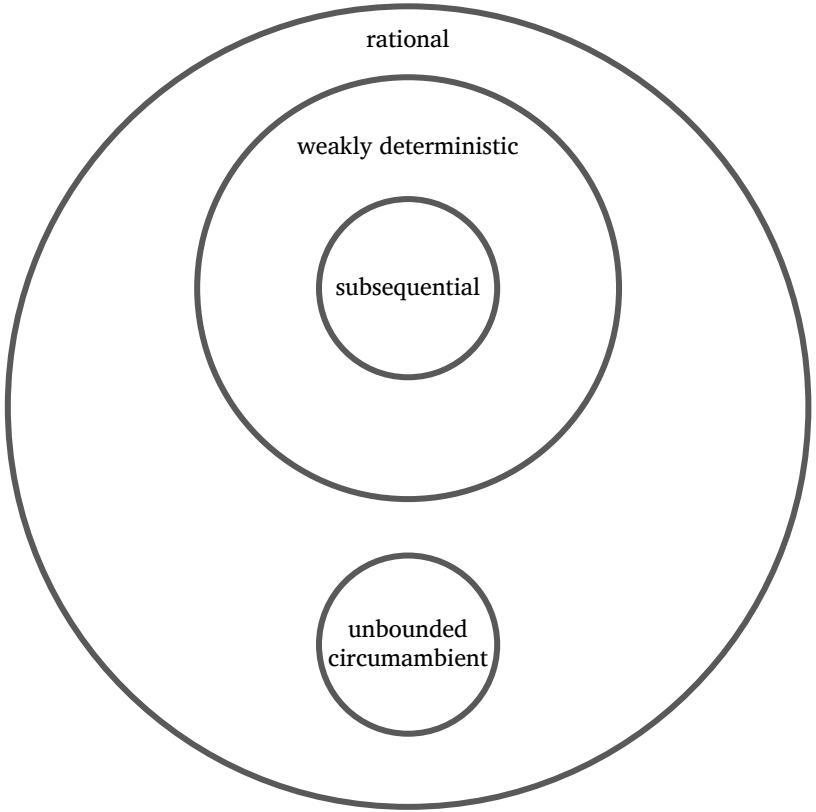
The *weakly deterministic* functions are a proposed proper subclass of the rational string functions (Heinz and Lai, 2013)

There are multiple definitions in the literature:

- Based on a composition of a L2R and R2L subsequential transducer, subject to special markup restrictions (Heinz and Lai, 2013)
- Based on a similar composition, but with restrictions based on *mutation maps* (Meinhardt et al., 2021)
- Based on a special subclass of bimachines (Meinhardt et al., 2024)
- Based on BMRS (Yolyan, 2025)

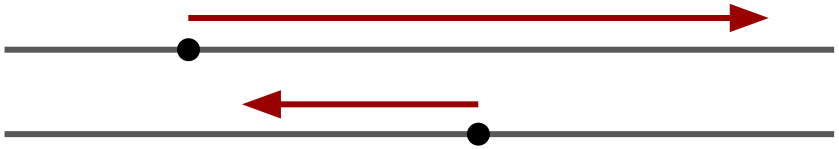
The above definitions do not necessarily define the same set of string functions

Weak determinism and unbounded circumambience



Meinhardt et al. (2024) draw the WD boundary around the *unbounded semiambient* maps.

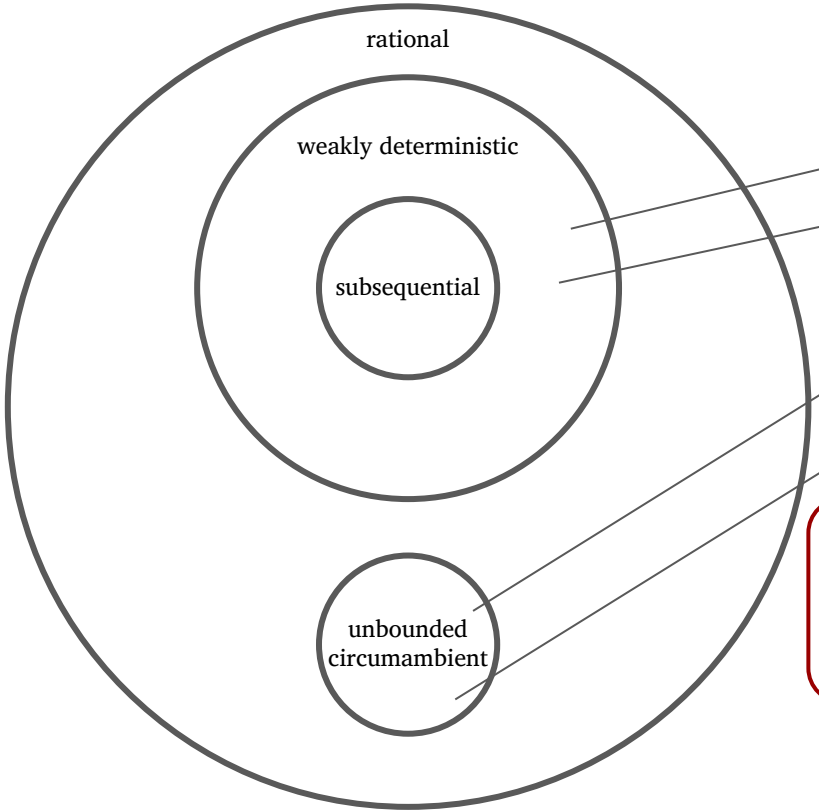
In unbounded semiambient maps, whether a segment undergoes alternation may be dependent on information arbitrarily far to its left or its right, but not both. The important side may differ for different segments in the string.



In unbounded circumambient maps (Jardine, 2016), the alternation of at least one segment necessarily depends on information from both sides, arbitrarily far away



Weak determinism and unbounded circumambience



Following (Yolyan, 2025):

- stem-controlled harmony
- LHOL stress
- unbounded tonal plateauing
- sour grapes

If we allow both L2R and R2L subsequential transducers, we can compute all of these phenomena, using the same technique in each case

LHOL Stress in Lushootseed

Leftmost heavy, otherwise leftmost stress (LHOL):

- Primary stress falls on the leftmost heavy syllable, if there is one
- If not, primary stress falls on the first syllable in the word
  - LHLHL → LH́LHL
  - LLLLL → ĹLLLL

LHOL Stress in Lushootseed

$\mathcal{R}$	L	□	L	□	□	□	□	□	□
$\mathcal{L}$	$\acute{L}$	$\acute{H}$	□	□	□	$\acute{L}$	□	□	□
In	L	H	L	H	L	L	L	L	L
Out	L	$\acute{H}$	L	H	L	$\acute{L}$	L	L	L

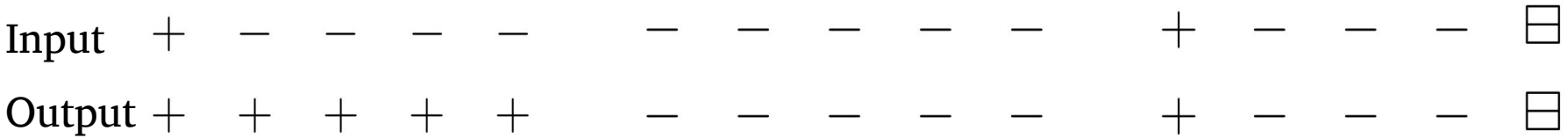
Unbounded tone plateauing in Luganda

In unbounded tonal plateauing, any L tone between two H tones at any distance will become high.

$\mathcal{R}$	□	□	□	□	L	□	□	L	L	L
$\mathcal{L}$	L	H	H	H	H	L	H	H	H	H
In	L	H	L	H	L	L	H	L	L	L
Out	L	H	H	H	L	L	H	L	L	L

Sour grapes

The sour grapes pattern describes a situation in which a harmony source (+) spreads its feature to targets (–) in one direction, unless a blocker (⊟) appears anywhere in that direction.



Sour grapes

$\mathcal{R}$	□	□	□	□	□	+	−	−	−	⊟	□	□	□	□	□
$\mathcal{L}$	+	+	+	+	+	+	+	+	+	+	□	□	□	□	□
In	+	−	−	−	−	+	−	−	−	⊟	−	−	−	−	−
Out	+	+	+	+	+	+	−	−	−	⊟	−	−	−	−	−

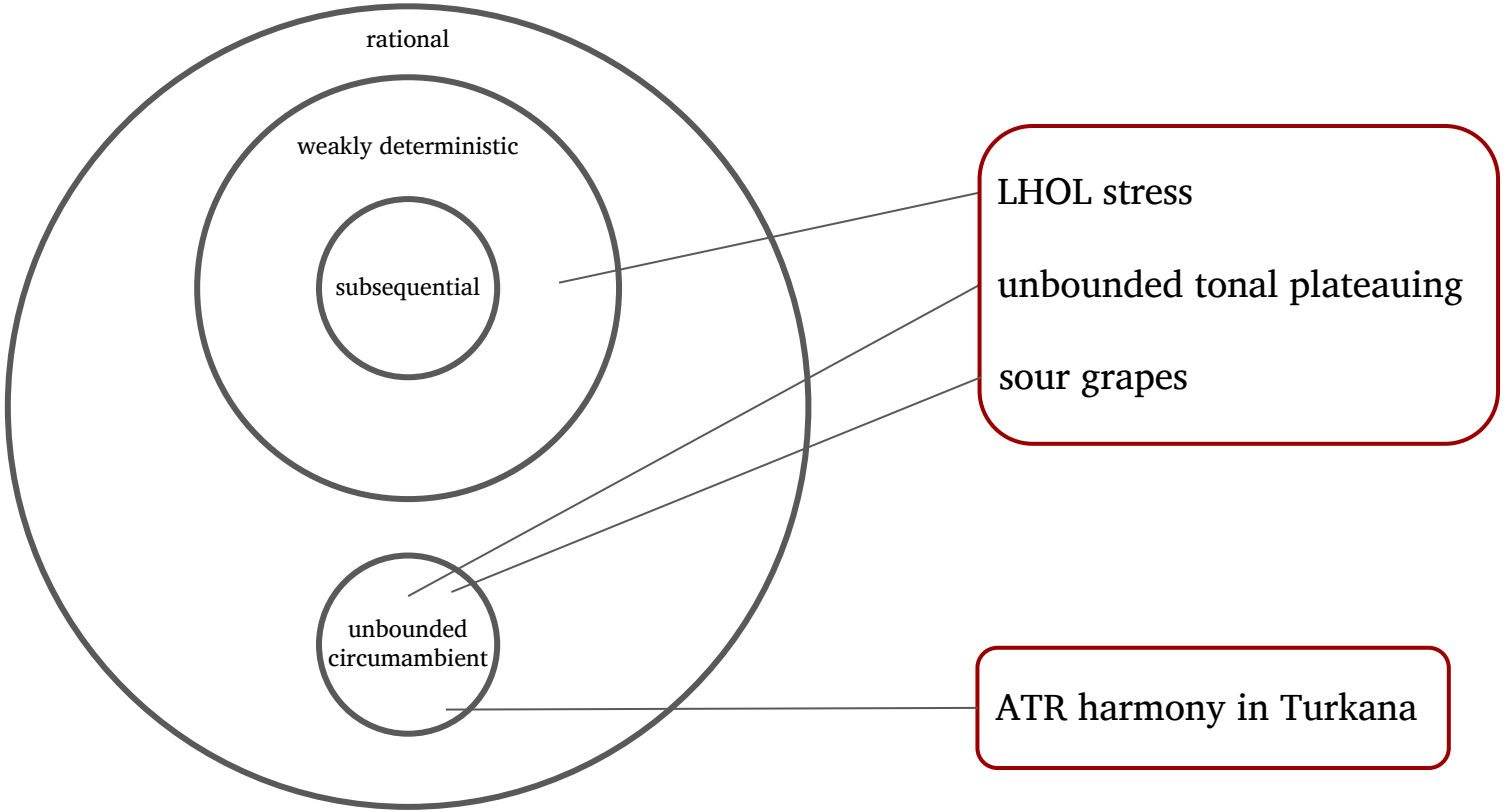
An interesting point: In each of these cases, we use *R* as a sort of ‘faithfulness transducer’

A segment needs information from its left only to determine what form it might change into, while it needs information from its right only to determine whether it may change at all

What can’t we do with this?*

*we think

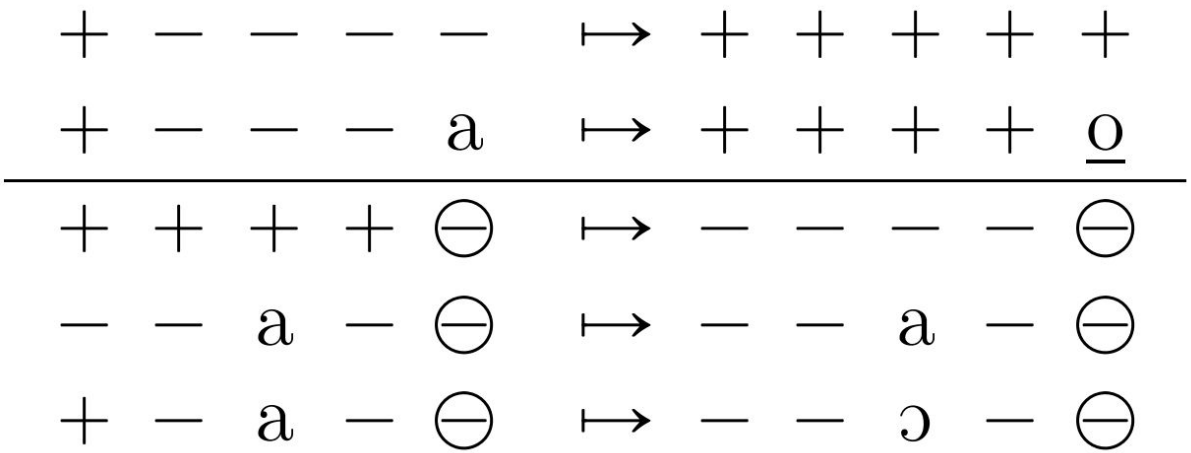
Weak determinism and unbounded circumambience



What's not possible? Turkana [ATR] harmony

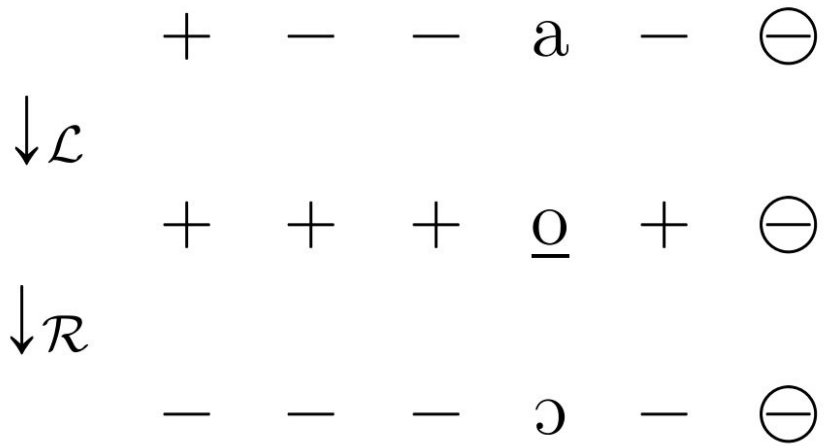
Here I factor away many of the details of Turkana [ATR] harmony

- We have L2R [+ATR] spread
- The low vowel [a] is repaired to [o̲] (+ATR)
- Exceptionally dominant [–ATR] vowels spread [–ATR] R2L
- When [a] occurs between a [+ATR] trigger and an exceptionally dominant [-ATR] trigger, it surfaces as [ɔ]



What's not possible? Turkana [ATR] harmony

This gives us a sense that the L2R function *must* feed the R2L function



This violates the basic idea of coproduct transductions, since each tier-transducer can only read the original input string.

Conclusion

- Coproduct transducers are intuitive to work with, and offer a different perspective on multiple-tier-based maps
- However, we currently have no idea how to formalize them, or to prove when something *cannot* be computed with them
- By allowing subsequential tier-transductions of opposite directions, we can compute certain WD and ND maps, without letting the machines “talk to each other” in any way, both operating only on the input.

Thank you!

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